

EEE2047S: Signals and Systems I

Class Test

28 September 2023

SOLUTIONS

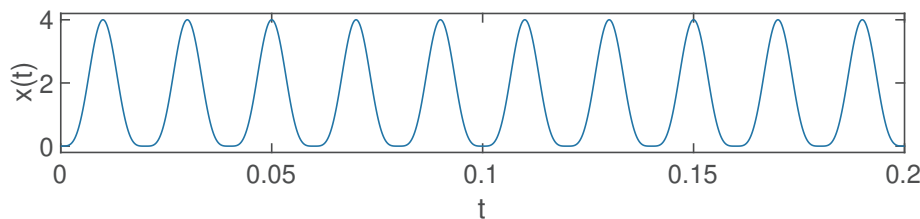
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Information

- The test is closed-book.
 - This test has *four* questions, totaling 20 marks.
 - There is an information sheet attached at the end of this paper.
 - Answer *all* the questions.
 - You have 60 minutes.
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1. (5 marks) The signal $x(t) = (\cos(100\pi t) - 1)^2$ is plotted below:



It can be expressed in the form $x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$ for some appropriate values of ω_0 and c_k .

- (a) Specify the fundamental frequency ω_0 .
 (b) Find the coefficients c_k and plot their magnitudes and phases over the range $k = -4$ to $k = 4$.

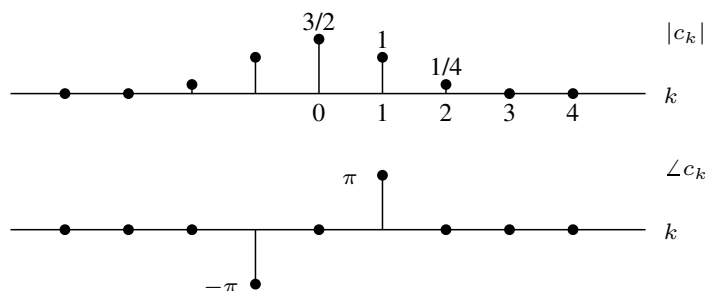
- (a) The fundamental period is $T = 0.1/5 = 0.02$ so $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{0.02} = 100\pi$ rad/s.
 (b) Multiplying out we can write

$$x(t) = \cos^2(100\pi t) - 2\cos(100\pi t) + 1.$$

Noting that $\cos^2(\theta) = 1/2 + 1/2 \cos(2\theta)$ this becomes

$$\begin{aligned} x(t) &= 3/2 - 2\cos(100\pi t) + 1/2 \cos(200\pi t) \\ &= 1/4 e^{-200\pi t} - e^{-100\pi t} + 3/2 - e^{100\pi t} + 1/4 e^{200\pi t}. \end{aligned}$$

Thus the series coefficients are $c_{-2} = c_2 = 1/4$, $c_{-1} = c_1 = -1$, and $c_0 = 3/2$, with all the others zero. Plot as follows:



2. (5 marks) A time signal $x(t)$ has Fourier transform $X(\omega) = 4/(4 + \omega^2)$. Find the Fourier transform of the following signals:

(a) $x_1(t) = x(2t - 3)$.

(b) $x_2(t) = \int_{-\infty}^t x(\tau) d\tau$.

(a) Suppose $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$ is a valid Fourier pair. Applying time scaling $y(at) \xleftrightarrow{\mathcal{F}} \frac{1}{a} Y\left(\frac{\omega}{a}\right)$ gives the new pair

$$x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{a} X\left(\frac{\omega}{a}\right).$$

Now applying time shift $y(t - c) \xleftrightarrow{\mathcal{F}} Y(\omega)e^{-j\omega c}$ to this pair gives the new pair

$$x(a(t - c)) \xleftrightarrow{\mathcal{F}} \frac{1}{a} X\left(\frac{\omega}{a}\right) e^{-j\omega c}.$$

For $a = 2$ and $c = 3/2$ this becomes

$$x(2t - 3/2) \xleftrightarrow{\mathcal{F}} \frac{1}{2} X\left(\frac{\omega}{2}\right) e^{-j\omega 3/2}.$$

Thus

$$X_1(\omega) = \frac{1}{2} \frac{4}{4 + (\omega/2)^2} e^{-j\omega 3/2}$$

(b) The integration property states that

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega} X(\omega) + \pi X(0) \delta(\omega),$$

so

$$X_2(\omega) = \frac{1}{j\omega} \frac{4}{4 + \omega^2} + \pi \frac{4}{4 + 0} \delta(\omega) = \frac{1}{j\omega} \frac{4}{4 + \omega^2} + \pi \delta(\omega).$$

3. (5 marks) Find the inverse Fourier transform of the signal

$$G(\omega) = \frac{1}{1+j\omega} \cos(2\omega) e^{-j5\omega}.$$

The given quantity can be written as

$$\begin{aligned} G(\omega) &= \frac{1}{1+j\omega} \frac{1}{2} e^{j2\omega} e^{-j5\omega} + \frac{1}{1+j\omega} \frac{1}{2} e^{-j2\omega} e^{-j5\omega} \\ &= \frac{1}{1+j\omega} \frac{1}{2} e^{-j3\omega} + \frac{1}{1+j\omega} \frac{1}{2} e^{-j7\omega}. \end{aligned}$$

Applying time shift to the pair

$$e^{-t} u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega + 1}$$

gives the new pair

$$e^{-(t-c)} u(t-c) \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega + 1} e^{-j\omega c}.$$

This can be used to invert each term in turn:

$$g(t) = \frac{1}{2} e^{-(t-3)} u(t-3) + \frac{1}{2} e^{-(t-7)} u(t-7).$$

4. (5 marks) Consider the continuous-time LTI system described by

$$\frac{dy(t)}{dt} + 2y(t) = x(t),$$

where $x(t)$ is the input and $y(t)$ is the output.

- (a) Using the Fourier transform, find the transfer function of the system.
- (b) Find the output $y(t)$ of the system for the input $x(t) = e^{-t}u(t)$.
- (c) Find the impulse response of the system.

- (a) The Fourier transform of the given expression is

$$j\omega Y(\omega) + 2Y(\omega) = X(\omega)$$

so the transfer function is

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{j\omega + 2}.$$

- (b) Since $X(\omega) = \frac{1}{j\omega + 1}$ the output (using partial fractions) is

$$Y(\omega) = H(\omega)X(\omega) = \frac{1}{j\omega + 2} \frac{1}{j\omega + 1} = \frac{1}{j\omega + 1} - \frac{1}{j\omega + 2}.$$

Thus

$$y(t) = e^{-t}u(t) - e^{-2t}u(t).$$

- (c) The impulse response is

$$h(t) = \mathcal{F}^{-1}\{H(\omega)\} = e^{-2t}u(t).$$

INFORMATION SHEET

Fourier transform properties

Property	Transform Pair/Property
Linearity	$ax(t) + bv(t) \iff aX(\omega) + bV(\omega)$
Time shift	$x(t - c) \iff X(\omega)e^{-j\omega c}$
Time scaling	$x(at) \iff \frac{1}{a}X(\frac{\omega}{a}) \quad a > 0$
Time reversal	$x(-t) \iff X(-\omega) = \overline{X(\omega)}$
Multiplication by power of t	$t^n x(t) \iff j^n \frac{d^n}{d\omega^n} X(\omega) \quad n = 1, 2, \dots$
Frequency shift	$x(t)e^{j\omega_0 t} \iff X(\omega - \omega_0) \quad \omega_0 \text{ real}$
Multiplication by $\cos(\omega_0 t)$	$x(t)\cos(\omega_0 t) \iff \frac{1}{2}[X(\omega + \omega_0) + X(\omega - \omega_0)]$
Differentiation in time domain	$\frac{d^n}{dt^n} x(t) \iff (j\omega)^n X(\omega) \quad n = 1, 2, \dots$
Integration	$\int_{-\infty}^t x(\lambda)d\lambda \iff \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$
Convolution in time domain	$x(t) * v(t) \iff X(\omega)V(\omega)$
Multiplication in time domain	$x(t)v(t) \iff \frac{1}{2\pi}X(\omega) * V(\omega)$
Parseval's theorem	$\int_{-\infty}^{\infty} x(t)v(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \overline{X(\omega)}V(\omega)d\omega$
Parseval's theorem (special case)	$\int_{-\infty}^{\infty} x^2(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) ^2 d\omega$
Duality	$X(t) \iff 2\pi x(-\omega)$

Common Fourier Transform Pairs

$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$	$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$
1 $(-\infty < t < \infty)$	$2\pi\delta(\omega)$
$-0.5 + u(t)$	$\frac{1}{j\omega}$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$\delta(t)$	1
$\delta(t - c)$	$e^{-j\omega c} \quad (c \text{ any real number})$
$e^{-bt}u(t)$	$\frac{1}{j\omega + b} \quad (b > 0)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0) \quad (\omega_0 \text{ any real number})$
$p_{\tau}(t)$	$\tau \text{sinc} \frac{\tau\omega}{2\pi}$
$\tau \text{sinc} \frac{\tau t}{2\pi}$	$2\pi p_{\tau}(\omega)$
$\left(1 - \frac{2 t }{\tau}\right) p_{\tau}(t)$	$\frac{\tau}{2} \text{sinc}^2 \left(\frac{\tau\omega}{4\pi}\right)$
$\frac{\tau}{2} \text{sinc}^2 \frac{\tau t}{4\pi}$	$2\pi \left(1 - \frac{2 \omega }{\tau}\right) p_{\tau}(\omega)$
$\cos(\omega_0 t + \theta)$	$\pi[e^{-j\theta}\delta(\omega + \omega_0) + e^{j\theta}\delta(\omega - \omega_0)]$
$\sin(\omega_0 t + \theta)$	$j\pi[e^{-j\theta}\delta(\omega + \omega_0) - e^{j\theta}\delta(\omega - \omega_0)]$
$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\frac{2\pi}{T})$

with $p_{\tau}(t) = u(t + \tau/2) - u(t - \tau/2)$ and $\text{sinc}(\lambda) = \sin(\pi\lambda)/(\pi\lambda)$.

Laplace transform properties

Property	Transform Pair/Property
Linearity	$ax(t) + bv(t) \iff aX(s) + bV(s)$
Time shift	$x(t-a)u(t-a) \iff e^{-as}X(s) \quad a \geq 0$
Time scaling	$x(at) \iff \frac{1}{a}X(\frac{s}{a}) \quad a > 0$
Frequency differentiation	$t^n x(t) \iff (-1)^n X^{(n)}(s)$
Frequency shift	$e^{at}x(t) \iff X(s-a)$
Differentiation	$x'(t) \iff sX(s) - x(0^-)$ $x''(t) \iff s^2X(s) - sx(0^-) - x'(0^-)$ $x^{(n)}(t) \iff s^n X(s) - s^{n-1}x(0^-) - \dots - x^{(n-1)}(0^-)$
Integration	$\int_{0^-}^t x(\lambda)d\lambda \iff \frac{1}{s}X(s)$ $\int_{-\infty}^t x(\lambda)d\lambda \iff \frac{1}{s}X(s) + \frac{1}{s} \int_{-\infty}^0 x(\lambda)d\lambda$
Time convolution	$x(t) * v(t) \iff X(s)V(s)$
Frequency convolution	$x(t)v(t) \iff \frac{1}{2\pi j}X(s) * V(s)$

Initial value: $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$

Final value: $f(\infty) = \lim_{s \rightarrow 0} sF(s)$ with all poles in left-hand plane

Common Unilateral Laplace Transform Pairs

$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st}ds$	$X(s) = \int_0^\infty x(t)e^{-st}dt$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{\lambda t}u(t)$	$\frac{1}{s-\lambda}$
$te^{\lambda t}u(t)$	$\frac{1}{(s-\lambda)^2}$
$t^n e^{\lambda t}u(t)$	$\frac{n!}{(s-\lambda)^{n+1}}$
$\cos(bt)u(t)$	$\frac{s}{s^2+b^2}$
$\sin(bt)u(t)$	$\frac{b}{s^2+b^2}$
$e^{-at} \cos(bt)u(t)$	$\frac{s+a}{(s+a)^2+b^2}$
$e^{-at} \sin(bt)u(t)$	$\frac{b}{(s+a)^2+b^2}$
$re^{-at} \cos(bt+\theta)u(t)$	$\frac{(r \cos \theta)s + (ar \cos \theta - br \sin \theta)}{s^2 + 2as + (a^2 + b^2)}$
$re^{-at} \cos(bt+\theta)u(t)$	$\frac{0.5re^{j\theta}}{s+a-jb} + \frac{0.5re^{-j\theta}}{s+a+jb}$
$re^{-at} \cos(bt+\theta)u(t)$	$\frac{As+B}{s^2+2as+c}$ $r = \sqrt{\frac{A^2c+B^2-2ABa}{c-a^2}}, b = \sqrt{c-a^2}, \theta = \tan^{-1} \frac{Aa-b}{A\sqrt{c-a^2}}$

Trigonometric identities

$$\begin{aligned} \sin(-\theta) &= -\sin(\theta) & \cos(-\theta) &= \cos(\theta) & \tan(-\theta) &= -\tan(\theta) & \sin^2(\theta) + \cos^2(\theta) &= 1 \\ \sin(2\theta) &= 2\sin(\theta)\cos(\theta) & \cos(2\theta) &= \cos^2(\theta) - \sin^2(\theta) = 2\cos^2(\theta) - 1 = 1 - 2\sin^2(\theta) \\ \sin(\theta_1 + \theta_2) &= \sin(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2) & \cos(\theta_1 + \theta_2) &= \cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2) \\ e^{j\theta} &= \cos(\theta) + j\sin(\theta) \end{aligned}$$